FRC SENSOR CAN — REVISION E REGULATED PROTOTYPE

Three fixed-threshold digital inputs. 6–22 V input; regulated 12 V and 5 V sensor power.
STM32F042 + TCAN3413. Digital status and edge counts over a custom team CAN protocol.
Not a CTRE Phoenix-compatible device. Inspect and bench-test prototypes before robot use.

Battery entry and internal supplies

File: power.kicad_sch

Regulated 12 V synchronous buck-boost

File: regulated_12v.kicad_sch

Protected sensor power rails

File: sensor_power.kicad_sch

Controller, clock, debug and status

File: controller.kicad_sch

CAN physical interface and termination

File: can.kicad_sch

Input 1: fixed-threshold digital input

File: input1.kicad_sch

Input 2: fixed-threshold digital input

File: input2.kicad_sch

Input 3: fixed-threshold digital input

File: input3.kicad_sch

Original design — custom WPILib CAN protocol

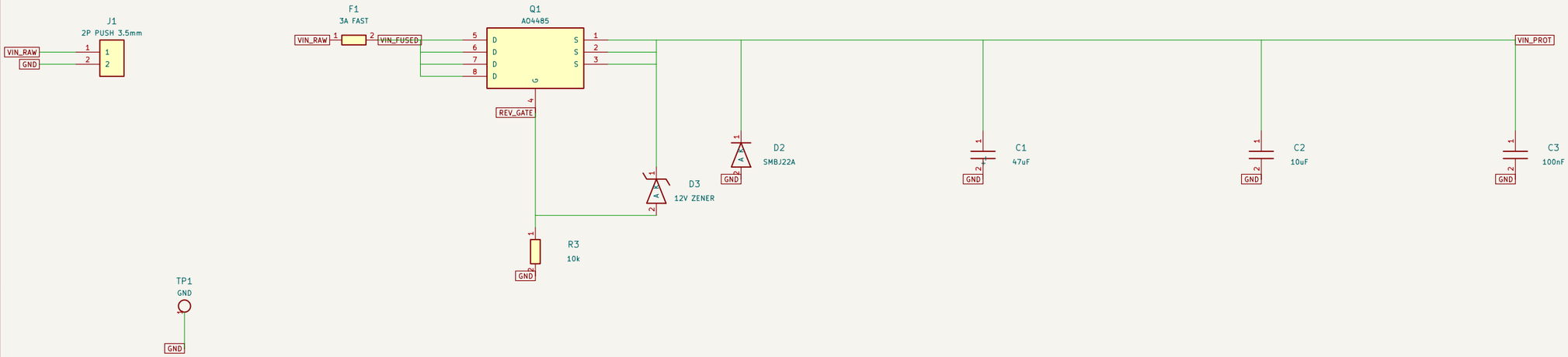
Sheet: /
File: FRC_Sensor_CAN.kicad_sch

Title: FRC Sensor CAN

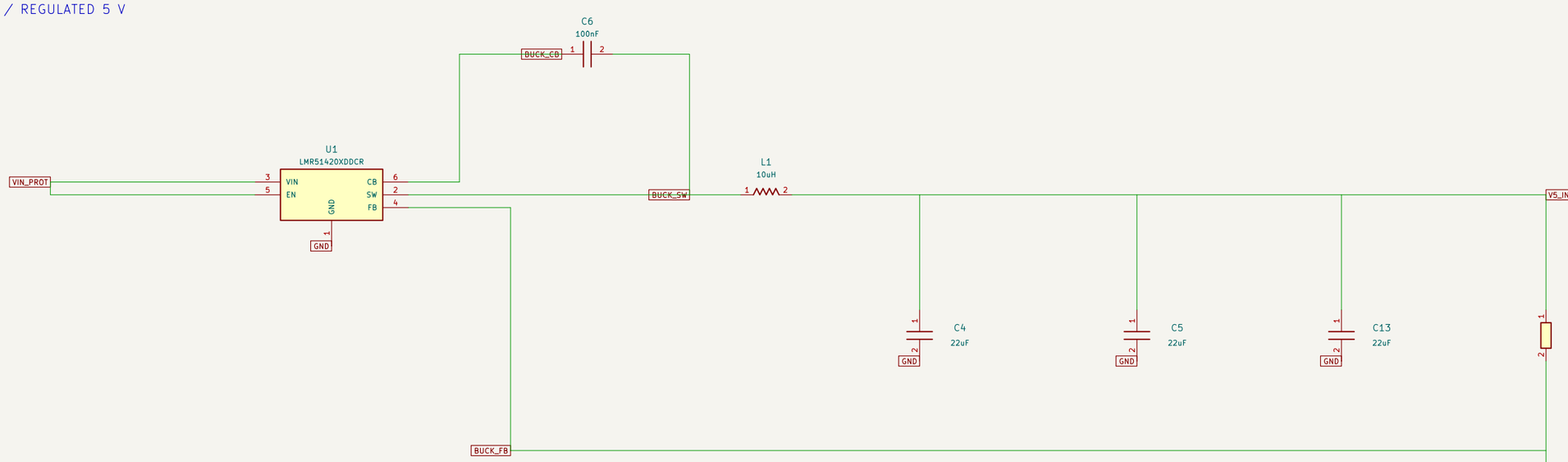
Size: A3	Date: 2026-09-08	**Rev: E**
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Battery entry and internal supplies

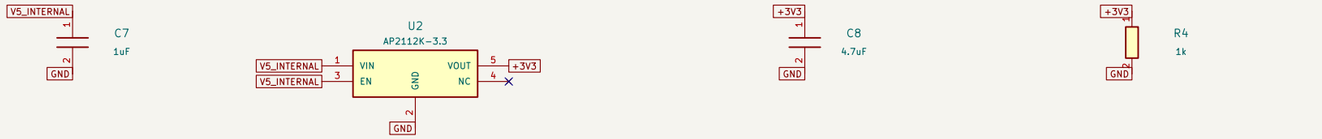
PROTECTED BATTERY ENTRY



500 kHz BUCK / REGULATED 5 V



3.3 V LOGIC SUPPLY



6–22 V input target. Single buck: 5.07 V nominal.
12 V buck-boost supply is on the next sheet.
R1/R2 = 102k / 13.7k, 0.1%. Buck-boost uses external bias from this 5 V rail.
TP1 is ground; use selectors/capacitor pads for rail checks.

Prototype: bench validation required before robot use

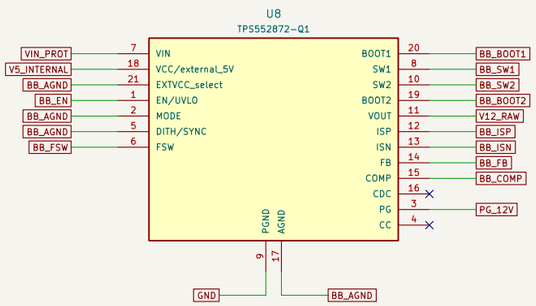
Original sensor interface design

Sheet: /Battery entry and internal supplies/
File: power.kicad_sch

Title: SC12 – Battery entry and internal supplies

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Regulated 12 V synchronous buck–boost





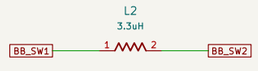








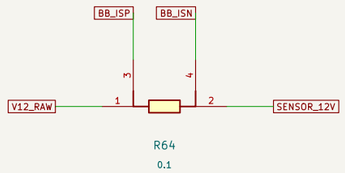






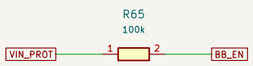


























12.011 V nominal: 300 mA useful shared sensor load. R64 sets approximately 0.5 A output current limit.
MODE and DITH grounded: automatic PFM for light–load efficiency. 24.9k sets 781 kHz nominal.
VCC supplied by the 5 V buck: EXTVCC is a LOGIC SELECT input tied low. No extra regulator.
Join AGND to PGND only at C68 negative pad using NT1. Keep ISP/ISN Kelvin traces separate from load copper.
PG goes to MCU PA3 with internal pull–up. Power–good is an indication, not a calibrated rail measurement.
Verify startup, current limiting, compensation, transients and temperature on assembled prototypes.

Prototype: bench validation required before robot use

Original sensor interface design

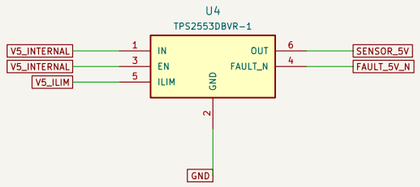
Sheet: /Regulated 12 V synchronous buck–boost/
File: regulated_12v.kicad_sch

Title: SC12 – Regulated 12 V synchronous buck–boost

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Protected sensor power rails

HARDWARE CURRENT LIMITING; NO FIRMWARE ENABLE









5 V sensor output: TPS2553-1 latch-off version, EN tied to input.
59k current-limit resistor. Cycle board power to clear overload.
FAULT_N goes to MCU PA4, with a firmware-enabled pull-up.
Design target: 300mA combined load on this rail, not per connector.
Do not back-power either sensor supply output.

Prototype: bench validation required before robot use
Original sensor interface design

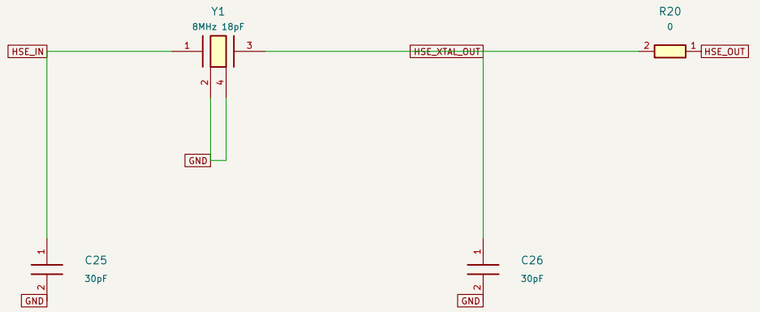
Sheet: /Protected sensor power rails/
File: sensor_power.kicad_sch

Title: SC12 – Protected sensor power rails

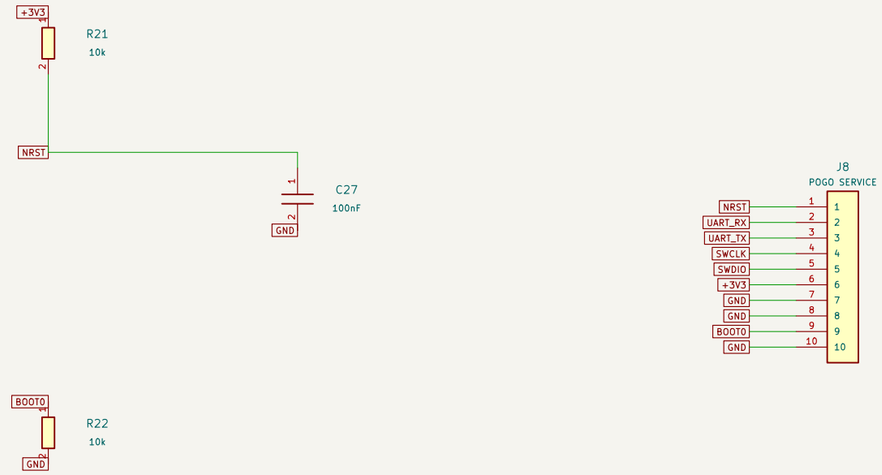
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Controller, clock, debug and status

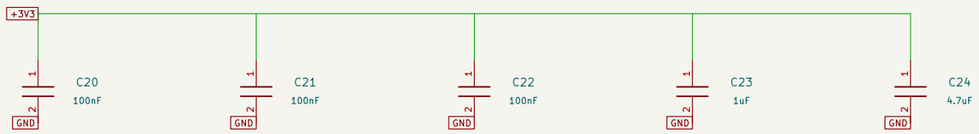
8 MHz CRYSTAL / PLL x6

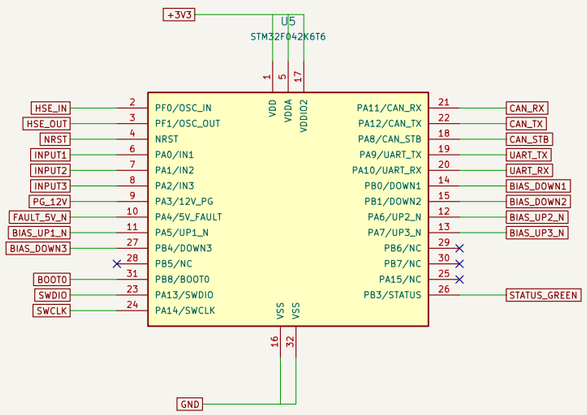


RESET / BOOT / SERVICE



LOCAL DECOUPLING





TC2050-IDC-NL mechanical footprint; CUSTOM electrical pinout.
Pin 6 is 3.3 V reference OUTPUT, never a programming power input.
Unused GPIOs are unconnected and initialized to analog mode.
PA3 monitors 12 V PG; PA4 monitors active-low 5 V fault. No rail ADCs.

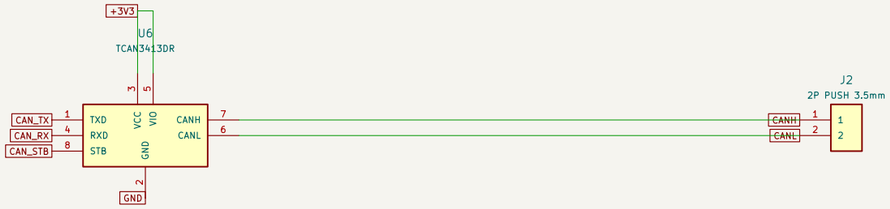
Prototype: bench validation required before robot use
Original sensor interface design

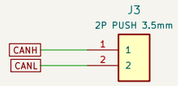
Sheet: /Controller, clock, debug and status/
File: controller.kicad_sch

Title: SC12 – Controller, clock, debug and status

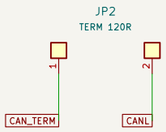
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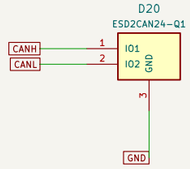
CAN physical interface and termination

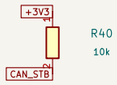


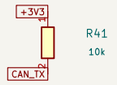




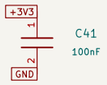






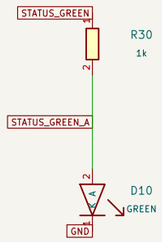






1 Mbit/s. Bridge JP2 only at a physical CAN trunk end.
TCAN3413 uses 3.3 V for both VCC and VIO.
STB pull-up keeps the bus silent during MCU reset.
CAN ground is shared through the power connection.

SINGLE STATUS INDICATOR



Prototype: bench validation required before robot use
Original sensor interface design

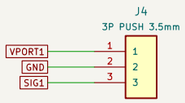
Sheet: /CAN physical interface and termination/
File: can.kicad_sch

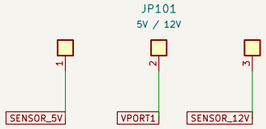
Title: SC12 – CAN physical interface and termination

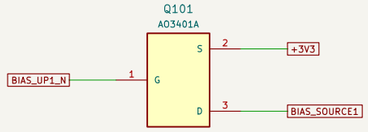
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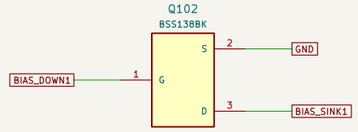
Input 1: fixed–threshold digital input

CAN–CONFIGURED INPUT BIAS / REMOVABLE SENSOR POWER

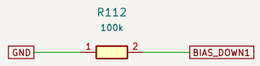


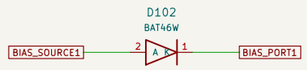


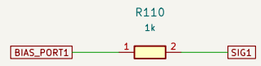






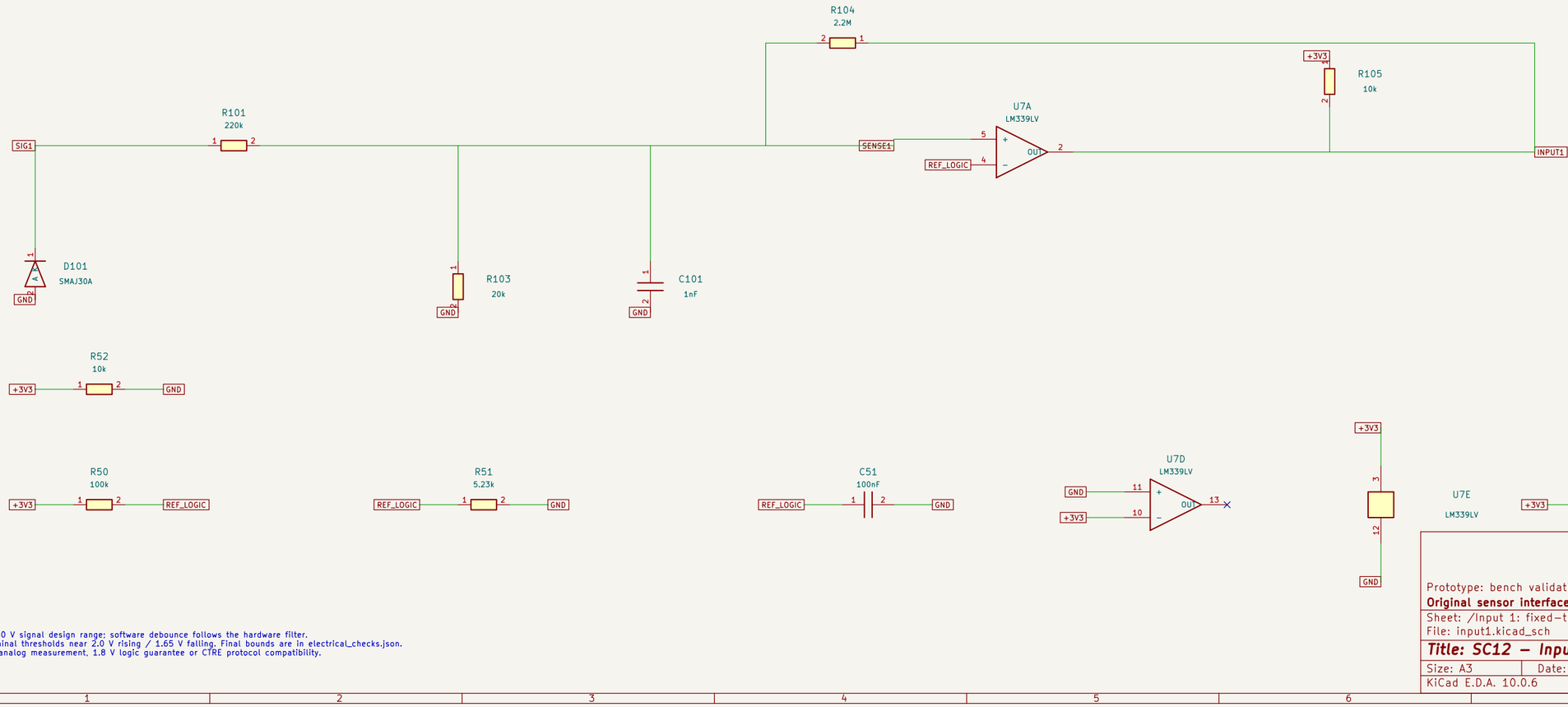








POWER: one 2 mm cap. 1–2=5 V or 2–3=12 V: no cap=off.
BIAS: 1k pull–up to internal 3.3 V through PMOS/diode; 10k pull–down through 60 V NMOS.
Both gates return OFF during reset. Software selects UP / DOWN / OFF / FLOAT / KEEPER. No bias bridges.



0–30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
Original sensor interface design

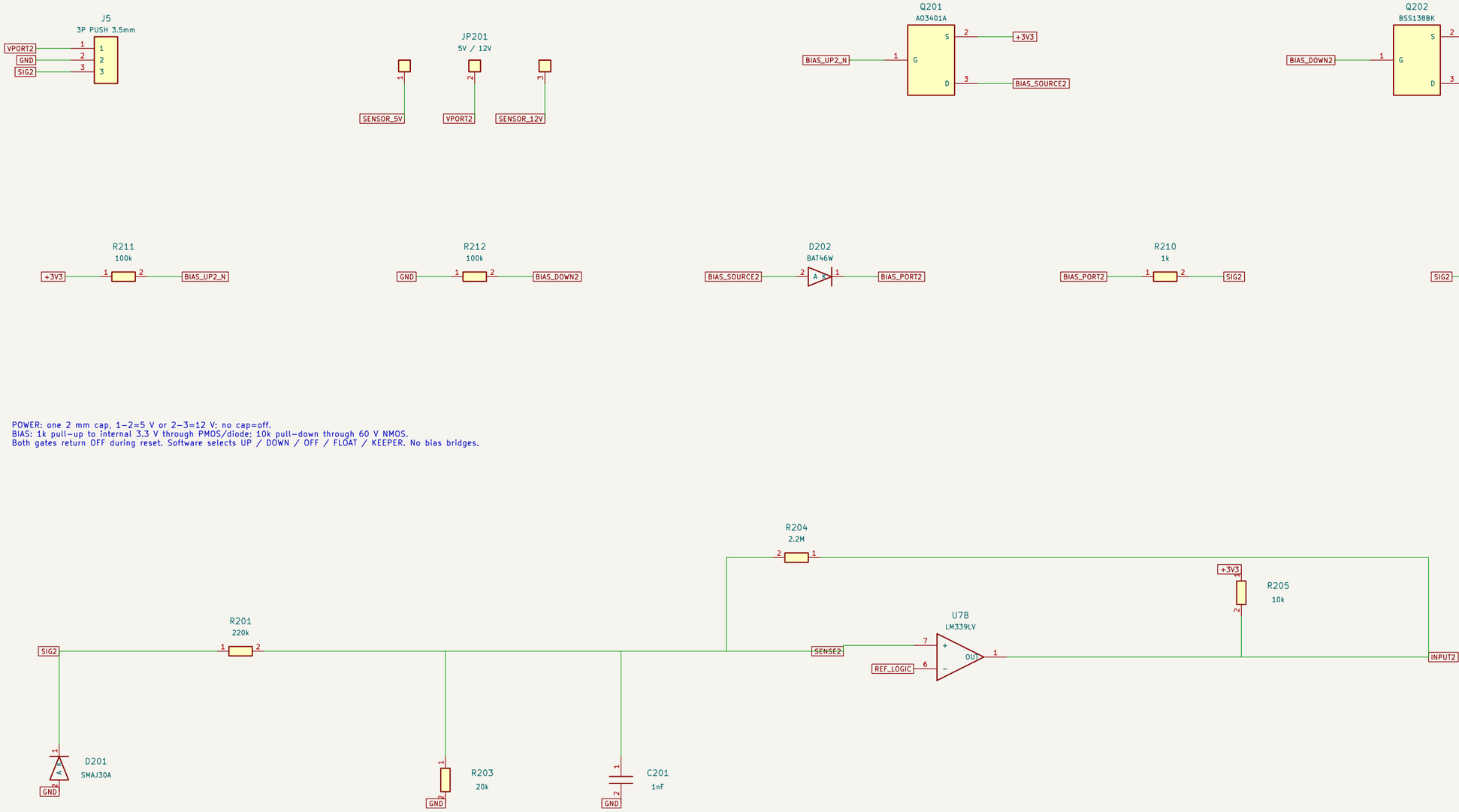
Sheet: /Input 1: fixed–threshold digital input/
File: input1.kicad_sch

Title: SC12 – Input 1: fixed–threshold digital input

Size: A3	Date: 2026–09–08	**Rev: E**
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Input 2: fixed-threshold digital input

CAN-CONFIGURED INPUT BIAS / REMOVABLE SENSOR POWER



POWER: one 2 mm cap. 1-2=5 V or 2-3=12 V: no cap=off.
BIAS: 1k pull-up to internal 3.3 V through PMOS/diode; 10k pull-down through 60 V NMOS.
Both gates return OFF during reset. Software selects UP / DOWN / OFF / FLOAT / KEEPER. No bias bridges.

The reference divider/filter and comparator power connections are on input1.
One fixed logic-compatible threshold for all three channels; no industrial threshold selector.

0-30 V signal design range; software debounce follows the hardware filter.
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Prototype: bench validation required before robot use
Original sensor interface design

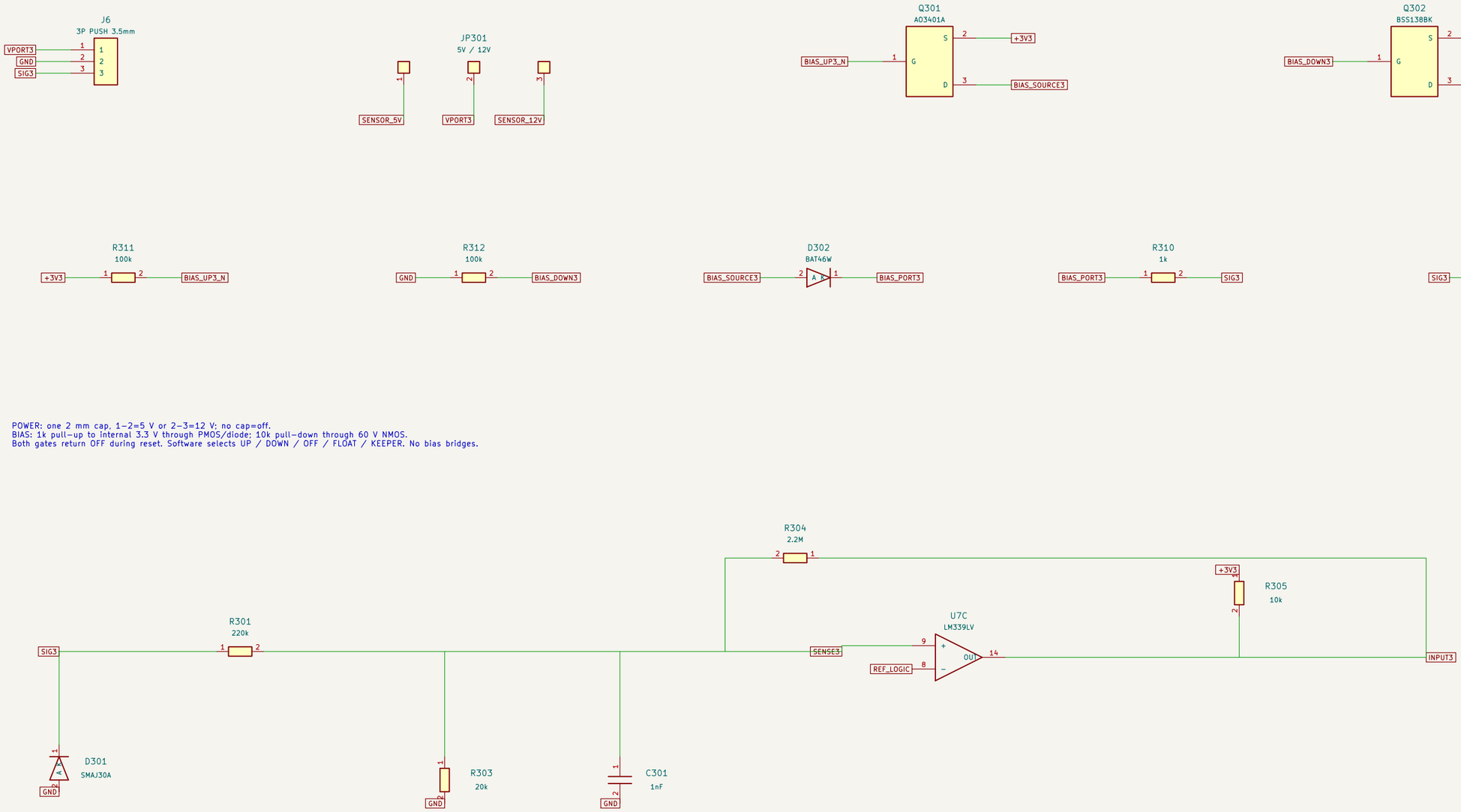
Sheet: /Input 2: fixed-threshold digital input/
File: input2.kicad_sch

Title: SC12 - Input 2: fixed-threshold digital input

Size: A3	Date: 2026-09-08	**Rev: E**
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Input 3: fixed–threshold digital input

CAN–CONFIGURED INPUT BIAS / REMOVABLE SENSOR POWER



POWER: one 2 mm cap. 1–2=5 V or 2–3=12 V: no cap=off.
BIAS: 1k pull–up to internal 3.3 V through PMOS/diode; 10k pull–down through 60 V NMOS.
Both gates return OFF during reset. Software selects UP / DOWN / OFF / FLOAT / KEEPER. No bias bridges.

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Prototype: bench validation required before robot use
Original sensor interface design

Sheet: /Input 3: fixed–threshold digital input/
File: input3.kicad_sch

Title: SC12 – Input 3: fixed–threshold digital input

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